

REST API Fundamentals

University Student Records API

Backend Development Project 1

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Introduction

This project was developed as part of Backend Development Project 1: REST API Fundamentals. The objective was to build a stateless web server capable of handling HTTP requests and returning structured JSON responses. The project demonstrates the implementation of RESTful API routes using Node.js and Express.js.

Project Objectives

- Create a local web server using Node.js and Express.js.
- Implement GET routes for retrieving data.
- Implement a POST route for adding new data.
- Return structured JSON responses.
- Test API endpoints using browser and Postman.
- Upload the completed project to GitHub.

Technologies Used

Technology	Purpose
Node.js	Runtime Environment
Express.js	Backend Framework
Postman	API Testing
Git	Version Control
GitHub	Repository Hosting

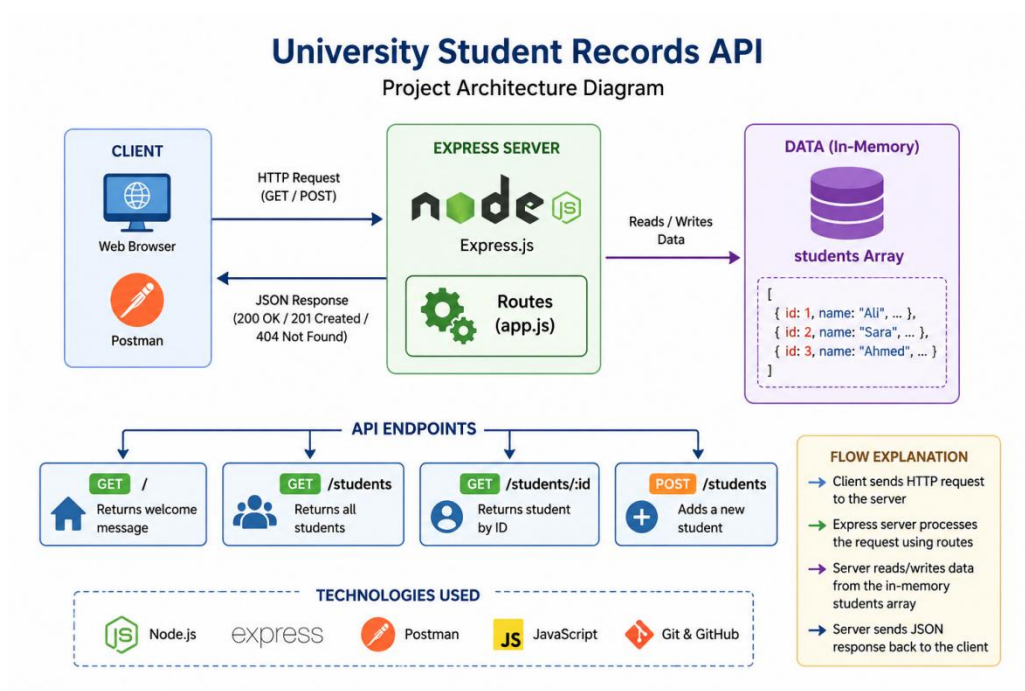
Project Structure

```
student-records-api
├── Documentation
├── Screenshots
├── node_modules
├── app.js
├── package.json
├── package-lock.json
├── README.md
└── .gitignore
```

API Endpoints

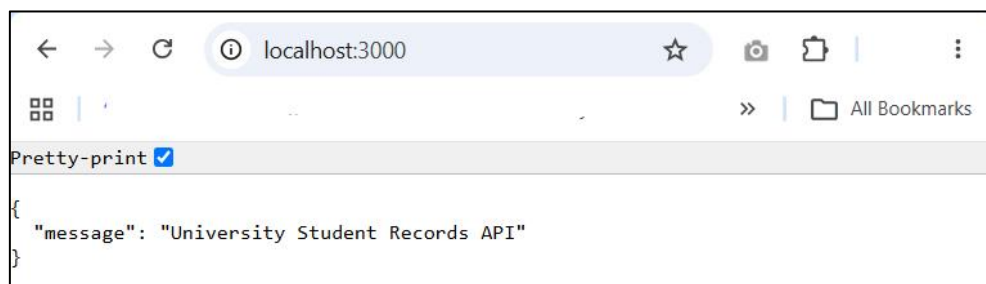
Method	Endpoint	Description
GET	/	Welcome Message
GET	/students	Get All Students
GET	/students/:id	Get Student By ID
POST	/students	Add New Student

Architecture Diagram

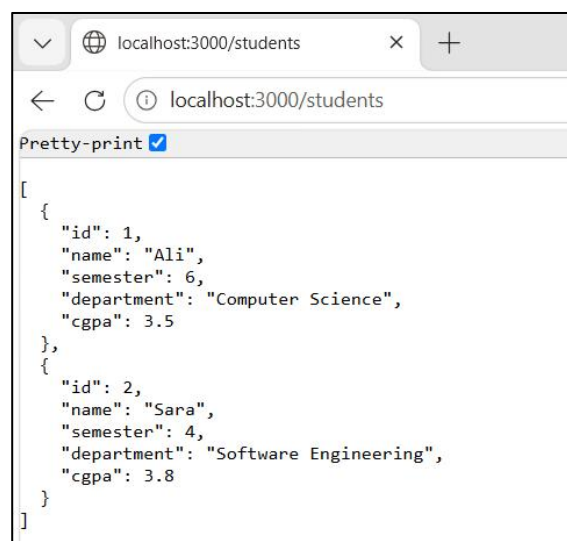


Testing and Results

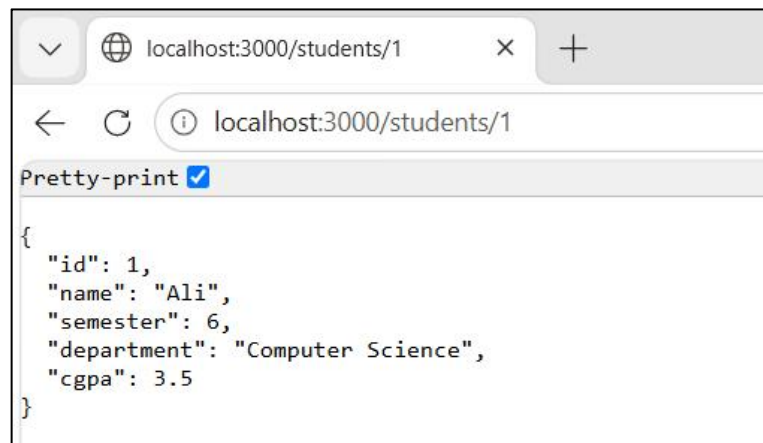
Home Route (/)



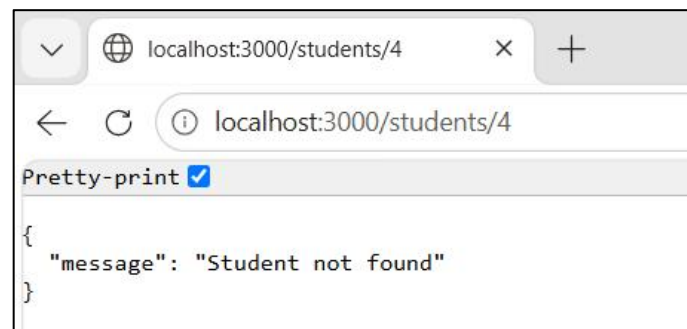
GET /students



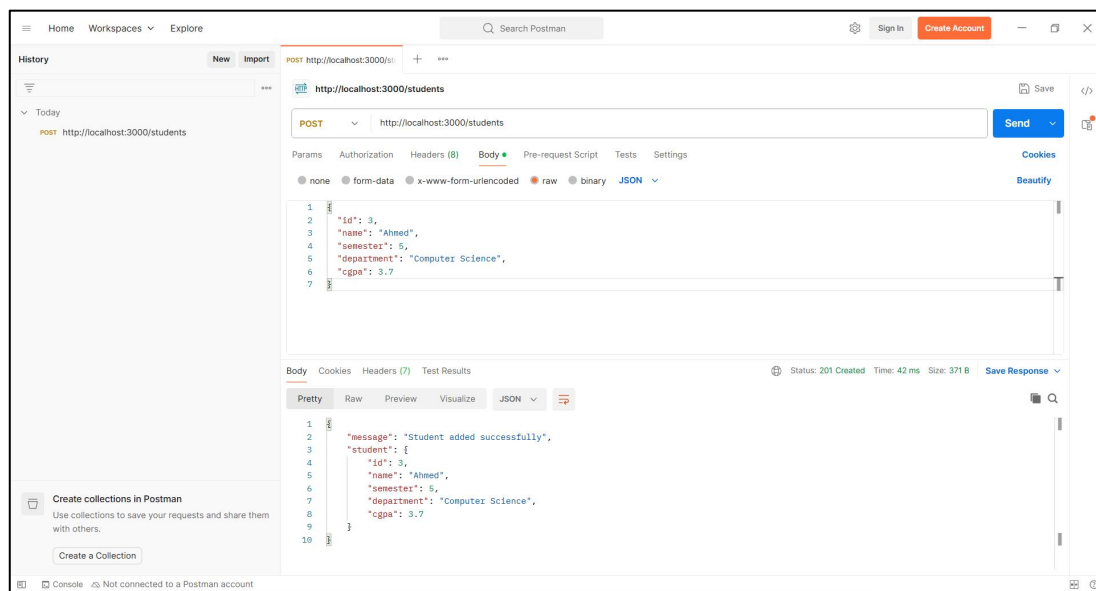
GET /students/1



GET /students/4



POST /students in Postman



GitHub Repository

<https://github.com/nosheenawan467/Task-1-Nosheen-Afzal>

Conclusion

The University Student Records API successfully demonstrates the core concepts of REST API Fundamentals. The project implements GET and POST routes, returns structured JSON responses, and follows stateless communication principles. Through this project, practical experience was gained in backend development using Node.js and Express.js.